Data-Model Hybrid Learning for Single Image Super Resolution

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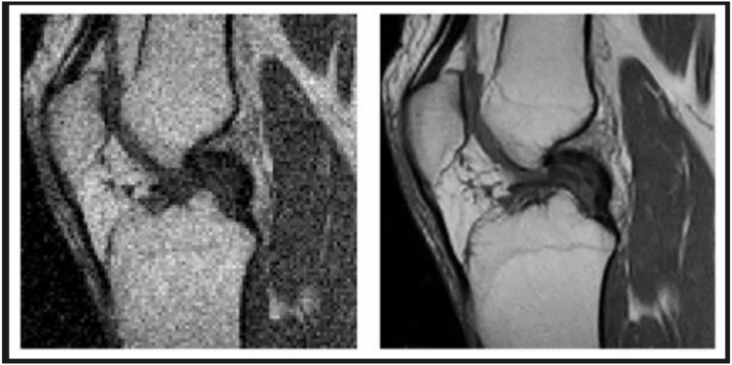
Outline

- Introduction of SISR
- Regularized Activation Functions
- Deep Image Prior
- Diffusion Model Performance and Its Variants
- Things We Have Tried: Adding regularity to different frameworks and in different parts (encoder, decoder, loss...)
- Future Work: DIP+Diffusion with spatial regularity, more problem adaptive priors

Single Image Super Resolution

LR: $\mathbf{y} \in \mathcal{R}^{m \times n \times p}$, HR: $\mathbf{x} \in \mathcal{R}^{M \times N \times P}$

$$\mathbf{y} = D(\mathbf{x}) + \epsilon$$



$$\mathrm{ReLU}(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

$$\mathrm{ReLU}(x) = argmin_{y \geq 0} \|x - y\|^2$$

We define the **STD-regularized ReLU** as:

$$ReLU_{STD}(x) = argmin_{y \geq 0} \|x - y\|^2 + \lambda \langle y, K_\sigma * (1 - y) \rangle$$

Linearization of the STD Term:

$R(y) \approx R(y^t) + \langle P(y^t), y - y^t \rangle, \quad \text{where } P(y^t) = K_\sigma * (1 - 2y^t) \in \partial R(y^t)$

The full objective becomes:

$$argmin_{y \geq 0} \|x - y\|^2 + \lambda \langle P(y^t), y \rangle$$

Thus, iterate $\mathrm{ReLU}_{STD}(x) = ReLU(x - \lambda P(y^t))$

STD Regularized Leaky ReLU

$$\text{LeakyReLU}(x) = \begin{cases} \alpha x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{LeakyReLU}(x) = \arg\min_{y} \left\{ \frac{1}{2}\|x - y\|^2 + \frac{1-\alpha}{2\alpha} \cdot \|\min(y, 0)\|^2 \right\}$$

We define the **STD-regularized Leaky ReLU** as:

$$\mathrm{LReLU}_{\mathrm{STD}}(x) = \arg\min_{y} \left\{ \frac{1}{2}\|x - y\|^2 + \frac{1-\alpha}{2\alpha} \cdot \|\min(y, 0)\|^2 + \lambda \langle y, K_\sigma * (1 - y) \rangle \right\}$$

Linearized Objective:

$$\min_{y} \left\{ \frac{1}{2}\|y - x\|^2 + \frac{1-\alpha}{2\alpha} \cdot \|\min(y, 0)\|^2 + \lambda \langle P(y^t), y \rangle \right\}$$

Deep Image Prior (DIP)[1]

They interpreted neural network as a parameterization of the image $x = f_\theta(z)$, where f denotes a network with its parameter θ (to be optimized) and z is a (fixed) vector of the same dimension as x.

Given the network architecture f, the parameter θ can be optimized via

$$\theta^* = \arg\min_\theta E(f_\theta(z); y) \Rightarrow x^* = f_{\theta^*}(z),$$

where E is a task-dependent data fidelity term and y is the degraded input.

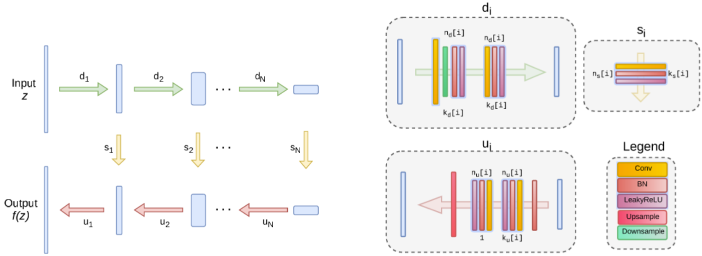
Specifically, for super-resolution, we consider the following model

$$E(x; y) = \|DBx - y\|_2^2,$$

where B is a blurring operator and D is a downsampling operator.

[1] Dmitry Ulyanov et al., CVPR, 2018

DIP rationales



- The network f_θ can capture some of the *low-level statistics* of natural images.
- The neural network reconstructs the clean image before converging to the corresponding degraded input, and consequently, an early stopping condition is desirable.

Motivation for STD Regularized DIP

- DIP does not fully consider the image characteristics, e.g., piecewise smoothness and edges.
- Embed STD regularized activation functions into the DIP architecture.

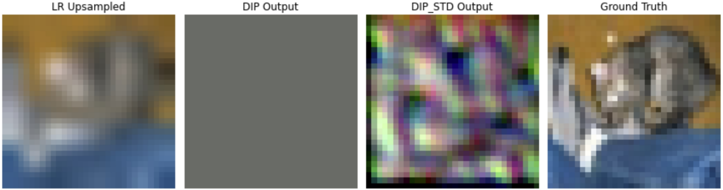


Figure: A toy DIP with 2 hidden layers, ReLu and STDReLu in the last layer.

- A limitation of DIP is that it processes only a single image. We aim to address this by integrating DIP with diffusion models.

Diffusion based SR

- DDPM (denoising diffusion probabilistic models)
  - **Forward**:

$$q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\sqrt{1-\alpha_t}\mathbf{z}_{t-1}, \alpha_t I) \tag{1}$$

    Here, α_t are schedules of adding noise, $\gamma_t = \Pi_{i=1}^{t}(1-\alpha_i)$

$$q(\mathbf{z}_0) \approx q(\text{HR images}) \tag{2}$$

  - **Backward**:

$$q(\mathbf{z}_T) \approx \mathcal{N}(0,1) \tag{3}$$

$$p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}(\mathbf{z}_{t-1}|\mu_\theta(\mathbf{z}_t, \gamma_t, \mathbf{x}), \Sigma_\theta(\mathbf{z}_t, \gamma_t, \mathbf{x})) \tag{4}$$

    which conditioned on LR image $\mathbf{x}$.
  - **Loss**

$$KL(q(\mathbf{z}_0, \cdots, \mathbf{z}_T)||p_\theta(\mathbf{z}_0, \cdots, \mathbf{z}_T)) \geq \mathbb{E}[-\log p_\theta(\mathbf{z}_0)] + c$$

$$\mathcal{L}_{DDPM}(\epsilon_\theta) := \mathbb{E}_{\substack{t\sim\mathcal{U}(1,T) \\ \mathbf{z}_0\sim q(\mathbf{z}_0) \\ \epsilon\sim\mathcal{N}(\mathbf{0},\mathbf{I})}}[\lambda(t)||\epsilon - \epsilon_\theta(\mathbf{x}, \mathbf{z}_t, \gamma_t)||^2] \tag{5}$$

TV+STDReLu

- Smooth Total Variation Loss — (differentiable version)

$$\mathcal{L}_{TV}(\mathbf{z}) = TV(u) = \sqrt{\sum_{i,j}(\mathbf{z}_{i+1,j} - \mathbf{z}_{i,j})^2 + (\mathbf{z}_{i,j+1} - \mathbf{z}_{i,j})^2 + \epsilon_{tv}} \quad (6)$$

- Total loss

$$\begin{aligned} \mathcal{L}(\theta) &= \mathcal{L}_{DDPM}(\epsilon_\theta) + \beta\mathcal{L}_{TV}(\hat{\mathbf{z}}_{0,\theta}) \\ \hat{\mathbf{z}}_{0,\theta} &:= \frac{1}{\gamma_t}[\mathbf{z}_t - \sqrt{1-\gamma_t}\epsilon_\theta(\mathbf{x}, \mathbf{z}_t, \gamma_t)] \end{aligned} \quad (7)$$

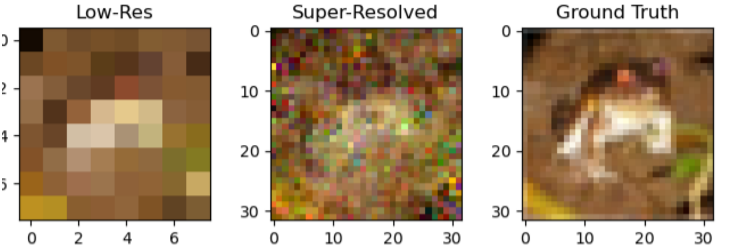


Figure: STDReLu+TV

	Standard DDPM	DDPM+TV	DDPM + STDRelu	DDPM + STDRelu +STDRelu+TV
PSNR	14.9238	16.2325	15.2827	16.3871
SSIM	0.2328	0.2057	0.1708	0.2164

TV+STDReLu

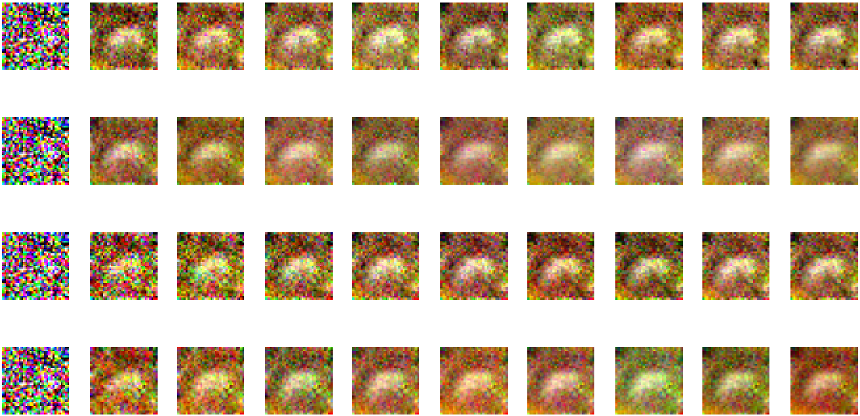
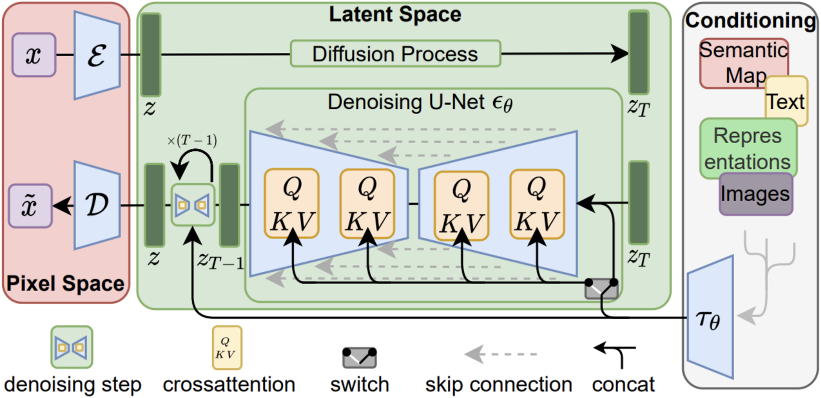


Figure: Denoising process v.s. epoch. Top Row: Standard DDPM; Second Row : DDPM+Total Variations; Third Row : DDPM+STDReLu; Bottom Row : DDPM+STDReLu+TV.

Latent diffusion model rationales



Unet leaky ReLU and STD leaky ReLU comparison

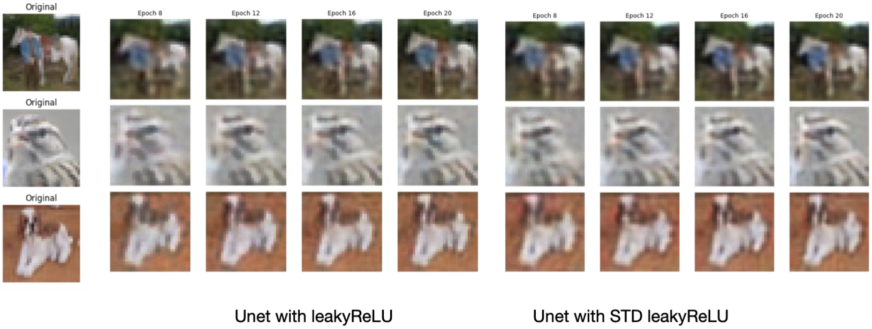


Figure: VAE reconstruction v.s. epochs for leaky ReLU and STD leaky ReLU as activation function

Loss for leaky ReLU: 37.0814
Loss for STD leaky ReLU: 34.7861

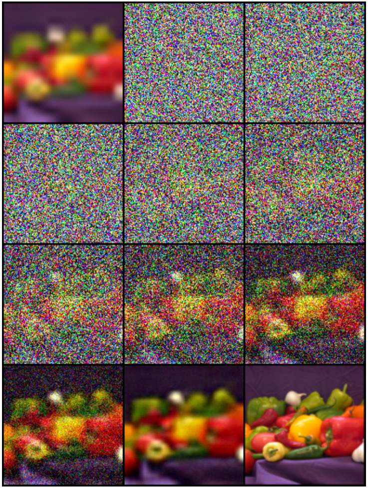
SR3 & DIP

- DIP
  - Implemented Image Super-Resolution using Deep Image Prior (DIP)
  - Used a randomly initialized CNN to reconstruct a high-resolution image from low-resolution input via implicit priors
  - Trained for 4000 iterations with LeakyReLU activation function for factor 8.
- SR3
  - Implemented Image Super Resolution via Iterative Refinement (SR3)
  - Adapted the code from Liangwei Jang[2]
  - Trained on CIFAR-10[3] data set (60k images) and tested on peppers image
  - 2000 diffusion time steps

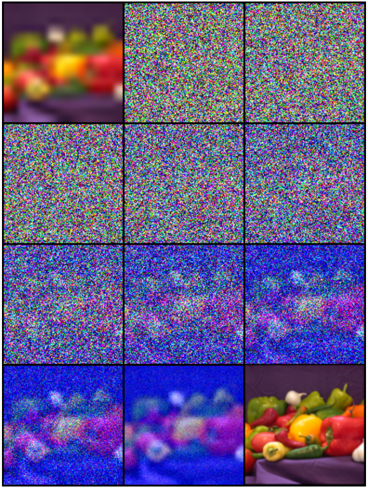
[2] https://github.com/Janspiry/Image-Super-Resolution-via-Iterative-Refinement

[3] A. Krizhevsky, G. Hinton, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

SR3 Progression



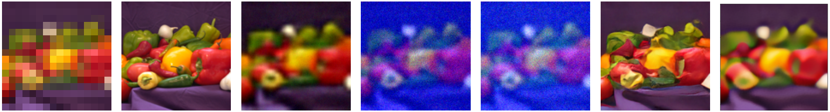
50k Iterations



100k Iterations

Figure: SR3 progression snapshots

Preliminary Results



LR | HR (gth) | 50k SR | 100k SR | Leaky SR | Pretr. SR | DIP SR

	SR3 (50k)	SR3 (100k)	SR3 (Pre-trained 640k)	SR3 leaky (10k)	DIP
PSNR	14.888	9.3052	17.072	9.3704	26.59
SSIM	0.33269	0.186	0.42931	0.18718	0.959

Future Work I: Regularized Diffusion Models

- Add regularization about the image to be recovered in the training loss to learn a better diffusion model.
- Study the DIP regularized with STD in the diffusion model.
- Adapt the regularized diffusion method to a state-of-the-art backbone architecture, e.g., SR3, and compare with other methods.

Future Work II: STD Regularized DIP+SR3

1. Build a DIP network (with STDReLU) to hallucinate a HR image that downscales to the input LR image
2. Sample noise and forward/backward steps using the diffusion process
3. Use DIP to generate a clean HR image from random noise
4. Feed DIP output to the diffusion model, e.g., SR3, to refine it conditioned on the LR image
5. Evaluate the quality of the recovered image via PSNR and SSIM